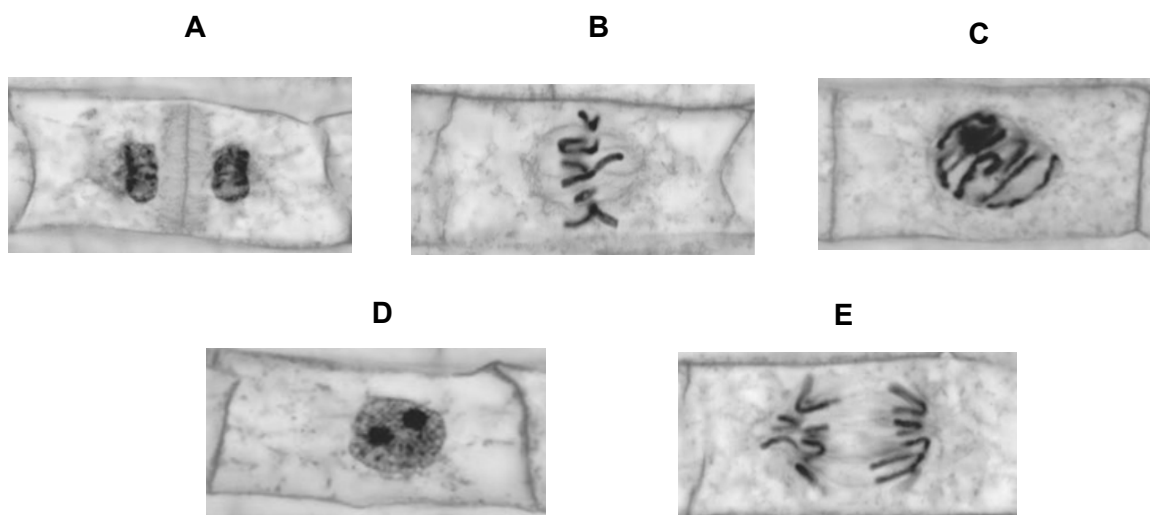


1 0 **Figure 8** shows some cells from an onion root tip at different stages of the cell cycle.

Figure 8



1 0 . 1 Place stages **A** to **E** in the correct order. Start with stage **D**.

[1 mark]

D

To obtain these images, the onion root tip was cut off, stained and put on a microscope slide. A cover slip was placed on top. The root tip was then firmly squashed and viewed under an optical microscope.

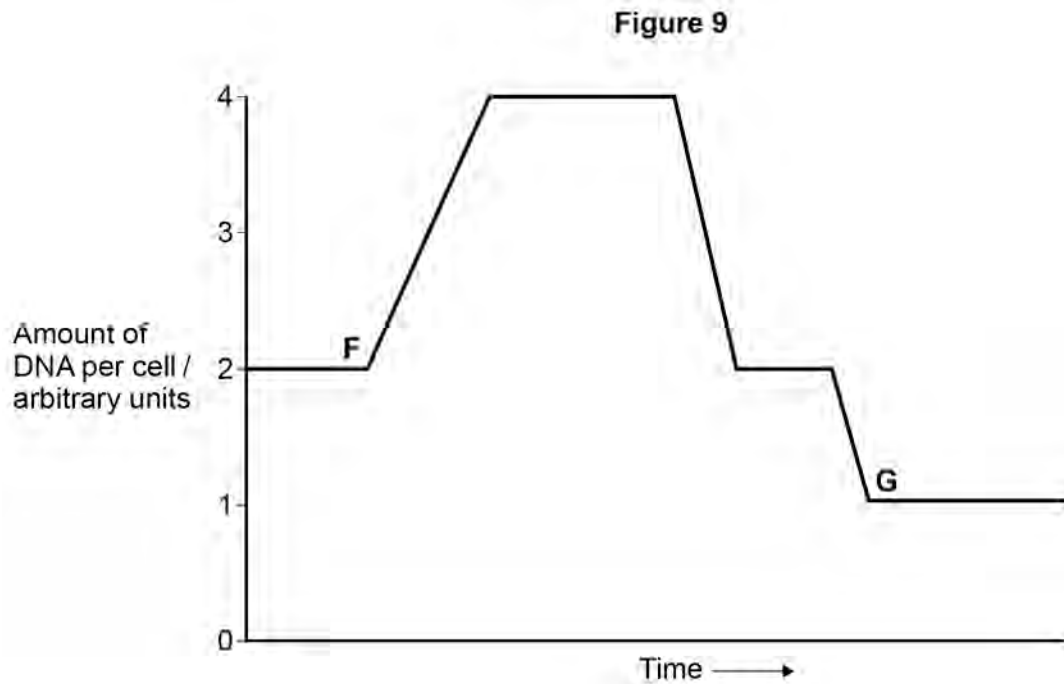
1 0 . 2 Complete **Table 5** to give **one** reason why each of these steps was necessary.

[2 marks]

Table 5

Step	Reason
Taking cells from the root tip	
Firmly squashing the root tip	

Figure 9 shows how the amount of DNA per cell changed during interphase and meiosis in an animal.



1 0 . 3

Explain how the behaviour of chromosomes causes these changes in the amount of DNA per cell between **F** and **G**.

[3 marks]

[Extra space]

1 0 . 4

What would happen to the amount of DNA per cell at fertilisation of cell **G**?

[1 mark]
